Implementation of change operations on workflow control-flow pattern based process models in BPMN language - Phase I & II

Pattern WCP-1 (Sequence)

Acceptance sequences ABC

Simple change operations

Insert
Since all activities have the same dependencies, we can consider the different cases for only one of the activities.
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $X \prec_d B$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $B \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
E8 Insert activity X, condition: $A \Leftrightarrow X; B \not\Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove
Remove activity A, B, C (all individual) - no problem
Modify
Modify ($A \Leftrightarrow B$), new dependency type ($A \Rightarrow B$) - possible (but does not lead to implication, but equivalence (dependency relaxation))
Modify ($A \Leftrightarrow B$), new dependency type ($A \not\Leftrightarrow B$) - possible
Modify ($A \Leftrightarrow B$), new dependency type ($A \overline{\wedge} B$) - possible
Modify ($A \Leftrightarrow B$), new dependency type ($A \vee B$) - possible (but does not lead to OR, but equivalence)
Modify ($A \Leftrightarrow B$), new dependency type ($A - B$) - possible (but does not lead to independence, but equivalence)

Modify ($A \prec_d B$), new dependency type ($A \prec B$) - possible (but does not lead to $\prec$, but $\prec_d$)
Modify ($A \prec_d B$), new dependency type ($A \succ_d B$) - possible (with console based application works; with BPC causes an error)
Modify ($A \prec C$), new dependency type ($A \prec_d C$) - not possible, no final matrix shown (seems to cause an error)

Combined change operations

Replace
Replace activity A with D - possible
Collapse
Collapse $\{A, B\}$ - possible
Collapse $\{A, B, C\}$ - possible
Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed
De collapse
Decollapse activity A - works for every provided YAML file

Parallelize
Parallelize $\{A, B\}$ - possible
Parallelize $\{B, C\}$ - possible
Parallelize $\{A, C\}$ - Error: Activities happening in between
Parallelize $\{A, B, C\}$ - possible

Swap
Swap A with B - possible
Swap A with C - possible
Skip
Skip A (also for all other activities) - possible
Condition Update
Condition activity: A, depending activity C - possible (also for all other combinations)

Pattern WCP-2 (Parallel Split)

Acceptance sequences ABC ACB BAC BCA CAB CBA

Simple change operations

Insert
Since all activities have the same dependencies, we can consider the different cases for only one of the activities.
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $X \prec B$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $X \prec_d B$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
E8 Insert activity X, condition: $A \Leftrightarrow X; B \not\Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove
Remove A (also or all other activities) - possible

Modify
For all existential dependencies we have the same dependency type ($a_i \Leftrightarrow a_j$), so we only perform modify for one pair of activities
Modify ($A \Leftrightarrow B$), new dependency type ($A \Rightarrow B$) - possible (but does not lead to implication, but equivalence (dependency relaxation))
Modify ($A \Leftrightarrow B$), new dependency type ($A \not\Leftrightarrow B$) - not possible, final result not displayed Modify ($A \Leftrightarrow B$), new dependency type ($A \overline{\wedge} B$) - not possible, final result not displayed
Modify ($A \Leftrightarrow B$), new dependency type ($A \vee B$) - possible (but does not lead to OR, but equivalence Modify ($A \Leftrightarrow B$), new dependency type ($A - B$) - possible (but does not lead to independency, but equivalence)

For all temporal dependencies we have the same dependency type ($a_i - a_j$), so we only perform modify for one pair of activities. Modify ($A - B$), new dependency type ($A \prec_d B$) - possible
Modify ($A - B$), new dependency type ($A \prec B$) - possible

Combined change operations

Replace
Replace activity A (also for any other activity) - possible

Collapse
Collapse $\{A, B\}$ - possible
Collapse $\{A, B, C\}$ - possible

De-collapse
De-collapse A (also for any other activity and any matrix) - possible

Parallelize
Parallelize $\{A, B\}$ - possible (the BPC application shows an error, but with console only tool it works)
Parallelize $\{A, B, C\}$ - possible

Swap
Swap A and B (for any two activities) - possible

Skip
Skip activity A (for any activity) - possible

Condition Update
Condition activity A, depending activity B (also for any other combination) - possible

Pattern WCP-3 (Synchronization)

Acceptance sequences ABC BAC

Simple change operations

Insert
Activities A, B have the same dependency to activity C, so we only need to consider one of the activities from the pair. For the combined constraints, consider combinations inside the pair and with activity C.
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
Insert activity X, condition: $X \prec_d C$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
Insert activity X, condition: $C \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec C$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec C$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $B \prec X$ - possible
Insert activity X, condition: $A \prec_d X$; $C \prec X$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $X \prec_d B$ - Error: The input is invalid: Temporal dependencies cause a contradiction.
Insert activity X, condition: $X \prec_d A$; $X \prec_d C$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $C \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
Insert activity X, condition: $C \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
Insert activity X, condition: $C \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
Insert activity X, condition: $X \Rightarrow C$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
Insert activity X, condition: $C \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
Insert activity X, condition: $C \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
Insert activity X, condition: $A \Leftrightarrow X; C \Leftrightarrow X$ - possible
E8 Insert activity X, condition: $A \Leftrightarrow X; B \not\Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.
Insert activity X, condition: $A \Leftrightarrow X; C \not\Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove
Remove activity A (also for any other activity) - possible

Modify
The dependency between A, B is the same pattern as WCP-2, so we only consider the dependencies between A or B and activity C.

Modify ($A \Leftrightarrow C$), new dependency type ($A \Rightarrow C$) - possible (but does not lead to implication, but equivalence (dependency relaxation))
Modify ($A \Leftrightarrow C$), new dependency type ($A \Leftarrow C$) - possible (but does not lead to implication, but equivalence (dependency relaxation))

Modify ($A \Leftrightarrow C$), new dependency type ($A \not\Leftrightarrow C$) - not possible, does not display a final result
Modify ($A \Leftrightarrow C$), new dependency type ($A \overline{\wedge} C$) - not possible, does not display a final result
Modify ($A \Leftrightarrow C$), new dependency type ($A \vee C$) - possible (but does not lead to or, but equivalence)
Modify ($A \Leftrightarrow C$), new dependency type ($A - C$) - possible (but does not lead to independence, but equivalence (dependency relaxation))

Modify ($A \prec C$), new dependency type ($A \prec_d C$) - possible
Modify ($A \prec C$), new dependency type ($A \succ C$) - possible, but leads to direct temporal dependency
Modify ($A \prec C$), new dependency type ($A - C$) - possible

Combined change operations

Replace
Replace activity A (also for any other activity) - possible

Collapse
Collapse $\{A, B\}$ - possible
Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed
Collapse $\{A, B, C\}$ - possible

De collapse
De-collapse A (also for any other activity and any matrix) - possible

Parallelize
Parallelize $\{A, B\}$ - possible
Collapse $\{A, C\}$ - Error
Collapse $\{A, B, C\}$ - possible

Swap
Swap activity A with activity B (also for any other pair of activities) - possible

Skip
Skip activity A (also for any other activity) - possible

Condition Update
Condition update, condition activity A, depending activity C - possible

Pattern WCP-4 (Exclusive Choice)

Acceptance sequences A B

Simple change operations

Insert
Since all activities have the same dependencies, we can consider the different cases for only one of the activities.
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $X \prec B$ - possible
T6 No contradiction possible

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \not\Leftrightarrow X$ - possible
E8 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove
Remove A - possible
Remove B - possible

Modify
Modify ($A \not\Leftrightarrow B$), new dependency type ($A \Leftrightarrow B$) - possible
Modify ($A \not\Leftrightarrow B$), new dependency type ($A \Leftarrow B$) - possible
Modify ($A \not\Leftrightarrow B$), new dependency type ($A \overline{\wedge} B$) - possible (but does not lead to NAND, but negated equivalence (dependency relaxation))
Modify ($A \not\Leftrightarrow B$), new dependency type ($A \vee B$) - possible
Modify ($A \not\Leftrightarrow B$), new dependency type ($A - B$) - possible (but does not lead to independence, but OR)

Modify ($A - B$), new dependency type ($A \prec B$) - possible (but does not lead to $\prec$, but independency)
Modify ($A - B$), new dependency type ($A \prec_d B$) - possible (but does not lead to $\prec$, but independency)

Combined change operations

Replace
Replace A with C - possible
Collapse
Collapse $\{A, B\}$ - possible
De collapse
De-collapse A and B (any process to decollapse) - possible
Parallelize
Parallelize $\{A, B\}$ - possible
Swap
Swap activity A and B - possible

Skip
Skip activity A (also for activity B) - possible
Condition Update
Condition update: condition activity A, depending activity B - possible

Pattern WCP-5 (Simple Merge)

Acceptance sequences AD BD CD

Simple change operations

Insert

Activities A, B, C have the same dependency to activity D, so we only need to consider one of the activities from the pair. For the combined constraints, consider combinations inside the pair and with activity C.

T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
Insert activity X, condition: $X \prec_d D$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
Insert activity X, condition: $D \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec D$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec D$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $B \prec_d X$ - possible
Insert activity X, condition: $A \prec_d X$; $X \prec D$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $D \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $D \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
Insert activity X, condition: $D \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
Insert activity X, condition: $D \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
Insert activity X, condition: $X \Rightarrow D$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
Insert activity X, condition: $D \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
Insert activity X, condition: $D \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
Insert activity X, condition: $A \Leftrightarrow X; D \Leftrightarrow X$ - possible
E8 No contradiction

Remove

Remove activity A (for any activity) - possible

Modify

Activities A, B, C have the same dependencies to each other, so we only consider one of the pairs. The dependencies to activity D are the same for A, B, C, so we only consider one of the pairs here.

Modify ($A \overline{\wedge} B$), new dependency type ($A \Leftrightarrow B$) - possible (but A, B no longer part of the result (contradiction))
Modify ($A \overline{\wedge} B$), new dependency type ($A \Rightarrow B$) - possible (but A no longer part of the result (contradiction))
Modify ($A \overline{\wedge} B$), new dependency type ($A \not\Leftrightarrow B$) - possible (but C no longer part of the result (contradiction))
Modify ($A \overline{\wedge} B$), new dependency type ($A \vee B$) - possible (but C no longer part of the result (contradiction))
Modify ($A \overline{\wedge} B$), new dependency type ($A - B$) - possible (but NAND instead of independence)

Modify $(A \Rightarrow D)$, new dependency type $(A \Leftrightarrow D)$ - possible (but B, C no longer part of the result (contradiction))
Modify $(A \Rightarrow D)$, new dependency type $(A \Leftarrow D)$ - possible (but B, C no longer part of the result (contradiction))
Modify $(A \Rightarrow D)$, new dependency type $(A \not\Leftrightarrow D)$ - possible
Modify $(A \Rightarrow D)$, new dependency type $(A \overline{\wedge} D)$ - possible (negated equivalence instead of NAND)
Modify $(A \Rightarrow D)$, new dependency type $(A \vee D)$ - possible
Modify $(A \Rightarrow D)$, new dependency type $(A - D)$ - possible (OR instead of independence)

Modify $(A - B)$, new dependency type $(A \prec B)$ - possible (but independence instead of $\prec$)
Modify $(A - B)$, new dependency type $(A \prec_d B)$ - possible (but independence instead of $\prec$)

Modify $(A \prec_d D)$, new dependency type $(A \prec D)$ - possible (but $\prec_d$ instead of $\prec$)
Modify $(A \prec_d D)$, new dependency type $(A \succ D)$ - possible (but $\succ_d$ instead of $\succ$)
Modify $(A \prec_d D)$, new dependency type $(A - D)$ - possible

Combined change operations

Replace
Replace activity A (also for any other activity) - possible
Collapse
Collapse $\{A, B\}$ - possible
Collapse $\{A, D\}$ - possible
Collapse $\{A, B, C\}$ - possible

De collapse
De-collapse A (for any other activity and any process to de-collapse) - possible

Parallelize
Parallelize $\{A, B\}$ - possible
Parallelize $\{A, D\}$ - possible
Parallelize $\{A, B, C, D\}$ - possible

Swap
Swap activity A and B (also for any pair of activities) - possible

Skip
Skip activity A - possible
Skip activity D - possible

Condition Update
Condition update, condition activity A, depending activity D - possible
Condition update, condition activity A, depending activity B - possible

Pattern WCP-6 (Multi-Choice)

Acceptance sequences AB AC AD ABC ABD ACB ACD ADB ADC ABCD ABDC ACBD ACDB ADBC ADCB

Simple change operations

Insert
Activities B, C, D have the same dependency to activity A, so we only need to consider one of the activities from the set. For the combined constraints, consider combinations inside the pair and with activity A.
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
Insert activity X, condition: $X \prec_d B$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
Insert activity X, condition: $B \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $X \prec B$ - possible
Insert activity X, condition: $B \prec_d X$; $C \prec X$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $B \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.
Insert activity X, condition: $X \prec_d C$; $X \prec_d B$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $B \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
Insert activity X, condition: $B \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
Insert activity X, condition: $B \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
Insert activity X, condition: $X \Rightarrow B$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
Insert activity X, condition: $B \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
Insert activity X, condition: $B \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
Insert activity X, condition: $B \Leftrightarrow X; C \not\Leftrightarrow X$ - possible
E8

Still open

Insert activity X, condition: $A \Leftrightarrow X; B \not\Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.
Insert activity X, condition: $A \Leftrightarrow X; C \not\Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove
Remove activity A (also for any other activity) - possible

Modify
Activities B, C, D all have the same dependencies to each other, so must only consider one pair of them. Additionally they share the same dependency to activity A, wo we only consider the pair (A, B).

Modify $(A \Rightarrow B)$, new dependency type $(A \Leftrightarrow B)$ - possible

Modify $(A \Leftarrow B)$, new dependency type $(A \Rightarrow B)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(Anot$
$LeftrightarrowB)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(A \overline{\wedge} B)$ - possible (instead of NAND we get negated equivalence)
Modify $(A \Rightarrow B)$, new dependency type $(A \vee B)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(A - B)$ - possible (instead of independence we get OR)

Modify $(B - C)$, new dependency type $(B \Leftrightarrow C)$ - possible
Modify $(B - C)$, new dependency type $(B \Rightarrow C)$ - possible (instead of implication we get equivalence (dependency relaxation))
Modify $(B - C)$, new dependency type $(B \not\Leftrightarrow C)$ - possible
Modify $(B - C)$, new dependency type $(B \overline{\wedge} C)$ - possible
Modify $(B - C)$, new dependency type $(B \vee C)$ - possible

Modify $(A \prec B)$, new dependency type $(A \prec_d B)$ - possible
Modify $(A \prec B)$, new dependency type $(A \succ B)$ - possible (direct temporal dependency instead of eventual)
Modify $(A \prec B)$, new dependency type $(A - B)$ - possible

Combined change operations

Replace
Replace activity A (also for any other activity) - possible

Collapse
Collapse $\{A, B\}$ - not possible, activities happening in between
Collapse $\{B, C\}$ - possible
Collapse $\{B, C, D\}$ - possible
Collapse $\{A, B, C, D\}$ - possible

De collapse
De-collapse activity A (also any other activity for any collapsed process) - possible

Parallelize
Parallelize $\{A, B\}$ - not possible, activities happening in between
Parallelize $\{B, C\}$ - possible
Parallelize $\{B, C, D\}$ - possible
Parallelize $\{A, B, C, D\}$ - possible

Swap
Swap activity A with B (or any other combination) - possible

Skip
Skip activity A (or any other activity) - possible

Condition Update
Condition update, condition activity A, depending activity B (or any other pair of activities) - possible

Pattern WCP-7 (Structured Synchronizing Merge)

Acceptance sequences AD BD CD ABD ACD BCD BAD CAD CBD ABCD ACBD BACD BCAD CABD CBAD

Simple change operations

Insert
Activities A, B, C have the same dependency to activity D, so we only need to consider one of the activities from the pair. For the combined constraints, consider combinations inside the pair and with activity C.
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
Insert activity X, condition: $X \prec_d D$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
Insert activity X, condition: $D \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec D$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec D$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $X \prec B$ - possible
Insert activity X, condition: $A \prec_d X$; $X \prec D$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $D \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $D \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
Insert activity X, condition: $D \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
Insert activity X, condition: $D \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
Insert activity X, condition: $X \Rightarrow D$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
Insert activity X, condition: $D \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
Insert activity X, condition: $D \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
Insert activity X, condition: $A \Leftrightarrow X; D \Leftrightarrow X$ - possible
E8 No contradiction

Remove
Remove activity A (also for any other activity) - possible

Modify
Activities A, B, C all have the same dependencies to each other, so must only consider one pair of them. Additionally they share the same dependency to activity D, wo we only consider the pair (A, B).

Modify $(C \Rightarrow D)$, new dependency type $(C \Leftrightarrow D)$ - possible
Modify $(C \Rightarrow D)$, new dependency type $(C \Leftarrow D)$ - possible
Modify $(C \Rightarrow D)$, new dependency type $(C \not\Leftrightarrow D)$ - possible
Modify $(C \Rightarrow D)$, new dependency type $(C \overline{\wedge} D)$ - possible (instead of NAND we get negated equivalence)
Modify $(C \Rightarrow D)$, new dependency type $(C \vee D)$ - possible
Modify $(C \Rightarrow D)$, new dependency type $(C - D)$ - possible (instead of independence we get OR)

Modify $(B - C)$, new dependency type $(B \Leftrightarrow C)$ - possible
Modify $(B - C)$, new dependency type $(B \Rightarrow C)$ - possible
Modify $(B - C)$, new dependency type $(B \not\Leftrightarrow C)$ - possible
Modify $(B - C)$, new dependency type $(B \overline{\wedge} C)$ - possible
Modify $(B - C)$, new dependency type $(B \vee C)$ - possible

Modify $(C \prec D)$, new dependency type $(C \prec_d D)$ - possible
Modify $(C \prec D)$, new dependency type $(C \succ D)$ - possible (direct temporal dependency instead of eventual)
Modify $(C \prec D)$, new dependency type $(C - D)$ - possible

Modify $(B - C)$, new dependency type $(B \prec C)$ - possible
Modify $(B - C)$, new dependency type $(B \prec_d C)$ - possible

Combined change operations

Replace
Replace activity A (also for any other activity) - possible

Collapse
Collapse $\{A, B\}$ - possible
Collapse $\{A, B, C\}$ - possible
Collapse $\{A, B, D\}$ - Error: Activity C happens between the activities to be collapsed
Collapse $\{A, B, C, D\}$ - possible

De collapse
De-collapse activity A (also for any other activity with any collapsed process) - possible

Parallelize
Parallelize $\{A, B, C\}$ - possible
Parallelize $\{A, B, D\}$ - Error
Parallelize $\{A, B, C, D\}$ - possible

Swap
Swap activity A with activity D (also for any pair of activities) - possible

Skip
Skip activity A (also for any other activity) - possible

Condition Update
Condition update, condition activity A, depending activity B (also for any other pair of activities)-possible

Pattern WCP-10 (Arbitrary Cycles)

Acceptance sequences ABC ABCBC ABCBCBC ... C BC BCBC BCBCBC ...

Simple change operations

Insert
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
Insert activity X, condition: $X \prec_d B$ - possible
Insert activity X, condition: $X \prec_d C$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
Insert activity X, condition: $B \prec_d X$ - possible
Insert activity X, condition: $C \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
Insert activity X, condition: $X \prec C$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
Insert activity X, condition: $X \prec C$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $B \prec X$ - possible
Insert activity X, condition: $A \prec_d X$; $X \prec B$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $B \prec X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $B \Leftrightarrow X$ - possible
Insert activity X, condition: $C \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
Insert activity X, condition: $B \not\Leftrightarrow X$ - possible
Insert activity X, condition: $C \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
Insert activity X, condition: $B \Rightarrow X$ - possible
Insert activity X, condition: $C \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
Insert activity X, condition: $X \Rightarrow B$ - possible
Insert activity X, condition: $X \Rightarrow C$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
Insert activity X, condition: $B \vee X$ - possible
Insert activity X, condition: $C \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
Insert activity X, condition: $B \overline{\wedge} X$ - possible
Insert activity X, condition: $C \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
E8 No contradiction

Remove
Remove activity A - possible Remove activity B - possible Remove activity C - possible
Modify
Modify $(A \Rightarrow B)$, new dependency type $(A \Leftrightarrow B)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(A \Leftarrow B)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(A \not\Leftrightarrow B)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(A \overline{\wedge} B)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(A \vee B)$ - possible
Modify $(A \Rightarrow B)$, new dependency type $(A - B)$ - possible

Modify $(A \Rightarrow C)$, new dependency type $(A \Leftrightarrow C)$ - possible
Modify $(A \Rightarrow C)$, new dependency type $(A \Leftarrow C)$ - possible
Modify $(A \Rightarrow C)$, new dependency type $(A \not\Leftrightarrow C)$ - possible (A no longer in the matrix (contradiction))
Modify $(A \Rightarrow C)$, new dependency type $(A \vee C)$ - possible

Modify $(B \Rightarrow C)$, new dependency type $(B \Leftrightarrow C)$ - possible
Modify $(B \Rightarrow C)$, new dependency type $(B \Leftarrow C)$ - possible
Modify $(B \Rightarrow C)$, new dependency type $(B \not\Leftrightarrow C)$ - possible (A no longer part of the process (contradiction)
Modify $(B \Rightarrow C)$, new dependency type $(B \barwedge C)$ - possible (negated existential equivalence instead of NAND, A no longer part of the process)
Modify $(B \Rightarrow C)$, new dependency type $(B - C)$ - possible (OR instead of independence)

Modify $(A \prec_d B)$, new dependency type $(A \prec B)$ - possible (direct temporal dependency instead of eventual)
Modify $(A \prec_d B)$, new dependency type $(A \succ B)$ - possible (A no longer part of the process (contradiction)
Modify $(A \prec_d B)$, new dependency type $(A - B)$ - possible (direct temporal dependency instead of independence)

Modify $(A \prec C)$, new dependency type $(A \prec_d C)$ - possible (A no longer part of the process (contradiction))
Modify $(A \prec C)$, new dependency type $(A \succ C)$ - possible (A no longer part of the process (contradiction)
Modify $(A \prec C)$, new dependency type $(A - C)$ - possible (eventual temporal dependency instead of independence)

Modify $(B \prec_d C)$, new dependency type $(B \prec C)$ - possible (direct instead of eventual)
Modify $(B \prec_d C)$, new dependency type $(B \succ C)$ - possible (A no longer part of the process (contradiction)
Modify $(B \prec_d C)$, new dependency type $(B - C)$ - possible

Combined change operations

Replace
Replace activity A (also possible for any other activity) with X - possible

Collapse
Collapse $\{A, B\}$ - possible
Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed
Collapse $\{B, C\}$ - possible
Collapse $\{A, B, C\}$ - possible

De collapse
De-collapse activity A (also for any other activity with any collapsed process) - possible

Parallelize
Parallelize $\{A, B\}$ - possible
Parallelize $\{A, C\}$ - Error: Activity B happens between the activities to be parallelized
Collapse $\{B, C\}$ - possible
Collapse $\{A, B, C\}$ - possible

Swap
Swap activity A with C (also for any other pair of activities) - possible

Skip
Skip activity A (also for any other activity) - possible

Condition Update
Condition update, condition activity A, depending activity B (also for any other pair of activities)-possible

Pattern WCP-17 (Interleaved Parallel Routing)

Acceptance sequences ABC ACB CAB

Simple change operations

Insert
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
Insert activity X, condition: $X \prec_d B$ - possible
Insert activity X, condition: $X \prec_d C$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
Insert activity X, condition: $B \prec_d X$ - possible
Insert activity X, condition: $C \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
Insert activity X, condition: $X \prec C$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
Insert activity X, condition: $X \prec C$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $B \prec X$ - possible
Insert activity X, condition: $A \prec_d X$; $X \prec B$ - possible
T6 Insert activity X, condition: $X \prec_d A$; $B \prec X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $B \Leftrightarrow X$ - possible
Insert activity X, condition: $C \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
Insert activity X, condition: $B \not\Leftrightarrow X$ - possible
Insert activity X, condition: $C \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
Insert activity X, condition: $B \Rightarrow X$ - possible
Insert activity X, condition: $C \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
Insert activity X, condition: $X \Rightarrow B$ - possible
Insert activity X, condition: $X \Rightarrow C$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
Insert activity X, condition: $B \vee X$ - possible
Insert activity X, condition: $C \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
Insert activity X, condition: $B \overline{\wedge} X$ - possible
Insert activity X, condition: $C \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
E8 Insert activity X, condition: $A \Leftrightarrow X, C \not\Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove
Remove activity A - possible
Remove activity B - possible
Remove activity C - possible

Modify
The pairs of activities (A, C) and (B, C) have the same dependencies, so we must only consider one of the pairs.

Modify $(A \Leftrightarrow B)$, new dependency type $(A \Rightarrow B)$ - possible (existential equivalence instead of

implication (dependency relaxation))
Modify ($A \Leftrightarrow B$), new dependency type ($A \Leftarrow B$) - possible (existential equivalence instead of implication (dependency relaxation))
Modify ($A \Leftrightarrow B$), new dependency type ($A \not\Leftrightarrow B$) - not possible, contradiction
Modify ($A \Leftrightarrow B$), new dependency type ($A \overline{\wedge} B$) - not possible, contradiction
Modify ($A \Leftrightarrow B$), new dependency type ($A \vee B$) - possible (existential equivalence instead of OR)
Modify ($A \Leftrightarrow B$), new dependency type ($A - B$) - possible (existential equivalence instead of independence)

Modify ($A \Leftrightarrow C$), new dependency type ($A \Rightarrow C$) - possible (existential equivalence instead of implication (dependency relaxation))
Modify ($A \Leftrightarrow C$), new dependency type ($A \Leftarrow C$) - possible (existential equivalence instead of implication (dependency relaxation))
Modify ($A \Leftrightarrow C$), new dependency type ($A \not\Leftrightarrow C$) - not possible, contradiction
Modify ($A \Leftrightarrow C$), new dependency type ($A \overline{\wedge} C$) - not possible, contradiction
Modify ($A \Leftrightarrow C$), new dependency type ($A \vee C$) - possible (existential equivalence instead of OR)
Modify ($A \Leftrightarrow C$), new dependency type ($A - C$) - possible (existential equivalence instead of independence)

Modify ($A \prec B$), new dependency type ($A \prec_d B$) - possible
Modify ($A \prec B$), new dependency type ($A \succ B$) - possible
Modify ($A \prec B$), new dependency type ($A - B$) - possible

Modify ($A - C$), new dependency type ($A \prec C$) - possible
Modify ($A - C$), new dependency type ($A \prec_d C$) - possible
Modify ($A - C$), new dependency type ($A \succ C$) - possible (direct temporal dependency instead of eventual)
Modify ($A - C$), new dependency type ($A \succ_d C$) - possible

Combined change operations

Replace

Replace activity A (also for any other activity) - possible

Collapse

Collapse $\{A, B\}$ - possible
Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed
Collapse $\{B, C\}$ - Error: Activity A happens between the activities to be collapsed
Collapse $\{A, B, C\}$ - possible

De collapse

De-collapse activity A (also for any other activity; with any collapsed process) - possible

Parallelize

Parallelize $\{A, B\}$ - possible (BPC application causes error, but works from algorithm logic and with console based application)
Parallelize $\{A, C\}$ - Error
Parallelize $\{B, C\}$ - Error
Parallelize $\{A, B, C\}$ - possible

Here must be an error in the software, collapse and parallelize should be possible / not possible for the same cases

Swap

Swap activity A with B (also for any other combination) - possible

Skip

Skip activity A - possible
Skip activity B - possible
Skip activity C - possible

Condition Update

Condition update, condition activity A, depending activity B - possible
Condition update, condition activity A, depending activity C - possible
Condition update, condition activity B, depending activity C - possible
Condition update, condition activity B, depending activity A - possible
Condition update, condition activity C, depending activity A - possible
Condition update, condition activity C, depending activity B - possible

Pattern WCP-19 (Cancel Activity)

Acceptance sequences AB A

Simple change operations

Insert
T0 Insert activity X, condition: none - possible
T1 Insert activity X, condition: $X \prec_d A$ - possible
Insert activity X, condition: $X \prec_d B$ - possible
T2 Insert activity X, condition: $A \prec_d X$ - possible
Insert activity X, condition: $B \prec_d X$ - possible
T3 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
T4 Insert activity X, condition: $X \prec A$ - possible
Insert activity X, condition: $X \prec B$ - possible
T5 Insert activity X, condition: $A \prec_d X$; $X \prec B$ - possible
T6 Insert activity X, condition: $A \prec_d X$, $B \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X, condition: none - possible
E1 Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $B \Leftrightarrow X$ - possible
E2 Insert activity X, condition: $A \not\Leftrightarrow X$ - possible
Insert activity X, condition: $B \not\Leftrightarrow X$ - possible
E3 Insert activity X, condition: $A \Rightarrow X$ - possible
Insert activity X, condition: $B \Rightarrow X$ - possible
E4 Insert activity X, condition: $X \Rightarrow A$ - possible
Insert activity X, condition: $X \Rightarrow B$ - possible
E5 Insert activity X, condition: $A \vee X$ - possible
Insert activity X, condition: $B \vee X$ - possible
E6 Insert activity X, condition: $A \overline{\wedge} X$ - possible
Insert activity X, condition: $B \overline{\wedge} X$ - possible
E7 Insert activity X, condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
E8 No contradiction

Remove
Remove activity A - possible
Remove activity B - possible

Modify
Modify $(A \Leftarrow B)$, new dependency type $(A \Leftrightarrow B)$ - possible
Modify $(A \Leftarrow B)$, new dependency type $(A \Rightarrow B)$ - possible
Modify $(A \Leftarrow B)$, new dependency type $(A \not\Leftrightarrow B)$ - possible
Modify $(A \Leftarrow B)$, new dependency type $(A \overline{\wedge} B)$ - possible (negated equivalence instead of NAND)
Modify $(A \Leftarrow B)$, new dependency type $(A \vee B)$ - possible
Modify $(A \Leftarrow B)$, new dependency type $(A - B)$ - possible

Modify $(A \prec_d B)$, new dependency type $(A \prec B)$ - possible (direct temporal dependency instead of eventual)
Modify $(A \prec_d B)$, new dependency type $(A \succ B)$ - possible (direct temporal dependency instead of eventual)
Modify $(A \prec_d B)$, new dependency type $(A \succ_d B)$ - possible
Modify $(A \prec_d B)$, new dependency type $(A - B)$ - possible

Combined change operations

Replace
Replace activity A - possible
Replace activity B - possible

Collapse
Collapse $\{A, B\}$ - possible

De collapse
De-collapse activity A (for any collapsed process) - possible
De-collapse activity B (for any collapsed process) - possible

Parallelize
Parallelize $\{A, B\}$ - possible

Swap
Swap activity A with B (or vice versa) - possible

Skip
Skip activity A - possible
Skip activity B - possible

Condition Update
Condition update, condition activity A, depending activity B - possible
Condition update, condition activity B, depending activity A - possible

Implementation of change operations for locked dependencies

Sequential process model We start with the most simple process model, a sequence of the activities A, B, C, D.
Locked dependency $A \prec_d B$
Delete activity A - possible
Delete activity C - possible
Insert activity X, condition: $C \prec X$ - possible
Collapse $\{A, B\}$ - possible
Collapse $\{B, C\}$ - possible
De-collapse A - possible
De-collapse C - possible
Replace A - possible
Replace C - possible
Skip activity A - possible
Skip activity C - possible
Swap activities A and B - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.
Swap activities A and C - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.
Swap activities C and D - possible Parallelize A and B - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.
Parallelize B and C - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.
Parallelize C and D - possible Condition update, A condition, B depending - possible
Condition update, A condition, C depending - possible
Condition update, C condition, D depending - possible

Locked dependency $A \Leftrightarrow B$
Delete activity A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.
Delete activity C - possible
Insert activity X, condition: $A \Leftrightarrow X$ - possible
Insert activity X, condition: $C \Leftrightarrow X$ - possible
Collapse $\{A, B\}$ - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.
Collapse $\{B, C\}$ - Error: Activity 'B' cannot be deleted because existential dependency from 'A' to 'B' is locked.
De-collapse A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.
De-collapse C - possible
Replace A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.
Replace C - possible
Skip activity A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.
Skip activity C - possible
Swap activities A and B - possible
Swap activities A and C possible
Swap activities C and D - possible
Parallelize A and B - possible
Parallelize B and C - possible
Parallelize C and D - possible Condition update, A condition, B depending - possible
Condition update, A condition, C depending - possible
Condition update, C condition, D depending - possible

XOR split We continue with the process with the acceptance sequences $\{AB, AC\}$. Here we focus on cases, which we did not consider beforehand.
Locked dependency $A \Leftarrow B$

Parallelize A, B, C - Error: Existential dependency from 'A' to 'B' is locked and cannot be changed.